

C S PATEL INSTITUTE OF TECHNOLOGY AND ENGINEERING - CHANGA

CIVIL ENGINEERING DEPARTMENT (FM & HYD. SECTION)

CL 372 FLUID MECHANICS – II

INDEX

Sr. No.	Date	Title	Page No.	Grade	Sign
1		Dimensional and model similitude			
2		Determination of viscosity by Capillary Tube			
3		Seepage Analysis through an earth dam by using viscous analogy			
4		To study compressible flow			
5		Determination of compressibility Factor (CCF) for pitot tube			
6		2D velocity profile of a turbulent flow in closed conduits			
7		Flow in closed conduits			
8		To study boundary layer flow			
9		Pressure distribution around cylinder			
10		To determine the lift force and lift coefficient of air foil			
11		To determine the coefficient of discharge of the standing wave venturiflume			
12		To determine discharge of the open channel with a slope			
13		To determine the conjugate depth and loss of specific energy in a hydraulic jump			

EXPERIMENT NO. 1

DIMENSIONAL AND MODEL SIMILITUDE

OBJECTIVE:

To understand the fundamentals of dimensional and model analysis

INTRODUCTION:

Dimensional Analysis:

Solving the practical problems in fluid mechanics usually requires both theoretical development and experimental results. By grouping significant quantities into dimensionless parameters, it is possible to reduce the number of variables

When the forces are acting on fluid flow, to make it non dimensional it is divided by the inertial force. The dimensions of fluid mechanics are force, mass, length and Time and they are related by Newton second law of motion. For compressible flow the thermal unit is to be considered. One common system employed in fluid mechanics is MLT system

The Buckingham π theorem is helpful in determination of non dimensional groups.

Model Similitude :

Model studies of proposed hydraulic structures and machines are frequently undertaken as aid to the designer. To have accurate results from the model study, there must be dynamic similitude between model and prototype. But this demands both geometric similitude and kinetic similitude between model and prototype.

Tutorial : Dimensional and Model Analysis

1. The pressure drop ΔP in flow of incompressible fluid through rough pipes is found to depend on the length l , average velocity u , fluid density, ρ , dynamic viscosity μ , diameter D and average roughness height e . Determine the dimensionless groups to correlate the flow parameters.

$$\frac{\Delta P}{\rho u^2} = f \left[\frac{L}{D}, \frac{e}{D}, \frac{\rho u D}{\mu} \right]$$

Ans :

2. The volume flow rate, Q over a V-notch depends on fluid properties namely density ρ , kinematic viscosity ν , and surface tension σ . It is also influenced by the angle of the notch, head of fluid over the vertex, and acceleration due to gravity. Determine the dimensionless parameters which can correlate the variables.

Ans :

$$Q = g^{1/2} h^{5/2} f \left[\frac{\nu}{g^{1/2} h^{3/2}}, \frac{\sigma}{\rho g h^2}, \theta \right]$$

3. State and explain the Buckingham's π Theorem

4. Describe the three types of similarities

5. List out the model laws and its applications

6. An aircraft fuselage has been designed for speeds of 380 kmph. To estimate power requirements the drag is to be determined. A model of 1/10 size is decided on. In order to reduce the effect of compressibility, the model is proposed to be tested at the same speed in a pressurized tunnel. Estimate the pressure required. If the drag on the model was measured as 100 N, predict drag on the prototype. **Ans : 1kN**

7. A scale model of a ship of $1/30$ size is to be towed through water. The ship is 135 m long. For similarity determine the speed with which the model should be towed. The ship is to travel at 30 kmph. **Ans : 1052m/s**

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EXPERIMENT NO. 2

DETERMINATION OF VISCOSITY BY CAPILLARY TUBE METHOD

OBJECTIVE:

To determine the viscosity of fluid by Capillary Tube method

APPARATUS:

Capillary tube is attached to tank (horizontal) as shown in Figure and liquid whose viscosity to be determined.

THEORY:

The Hagen Poiseuille equation gives discharge through a small diameter circular tube in laminar flow

Is given as

$$Q = \frac{\pi d^4 (p_1 - p_2)}{128 \mu L} \text{ and rearranging which gives}$$

$$\mu = \frac{\pi d^4 (p_1 - p_2)}{128 Q L}$$

It is necessary to measure the pressure difference between two points which must lie in the region of fully developed laminar flow. The pressure difference is also known in terms of h since $p_2 = 0$

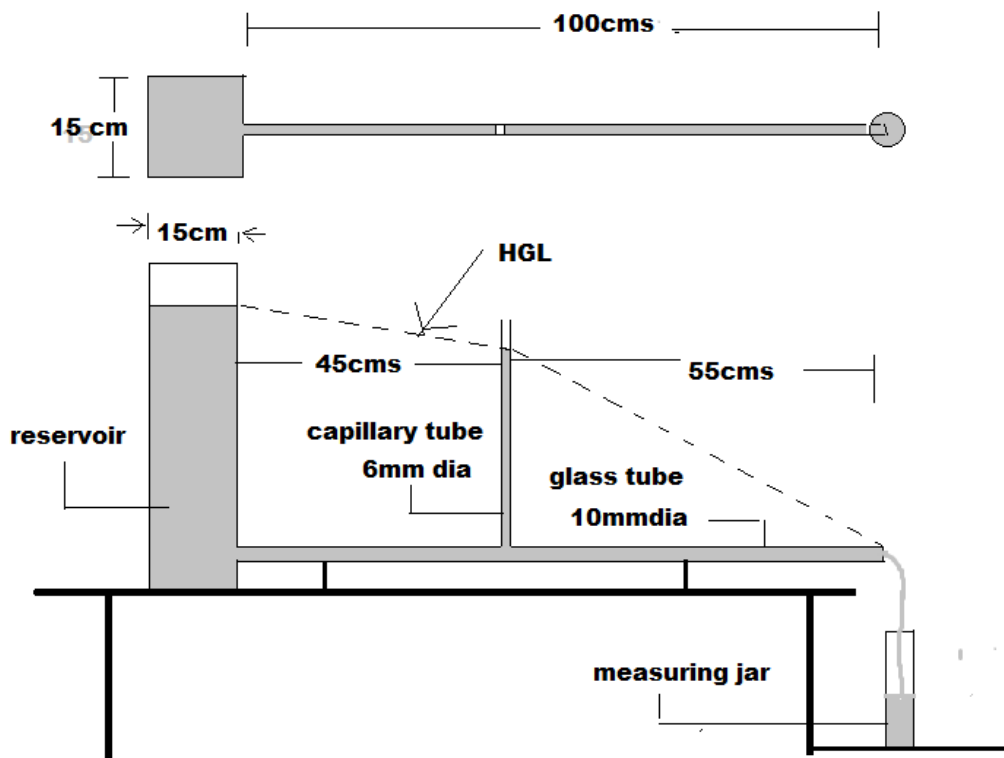
$$p_1 - p_2 = \gamma h$$

PRACTICAL SIGNIFICANCE:

Capillary tube viscometer is suitable for low viscous fluid and also used to measure viscosity of blood and pulp of paper. However viscosity is proportional to 4th power of diameter, the variation in viscosity may be about 0.5 %.

PROCEDURE:

Fill the tank with fluid about 2/3 of capacity. Let the flow pass through the capillary tube whose diameter is measured by caliper. In the beginning the height is H_1 and at the end of experiment the Height measured is H_2 . So H is $(H_1 + H_2) / 2$. Collect the fluid in measuring cylinder for known time and find Q . Lastly find the viscosity of fluid. Take observations for 5 different discharges.



FIXED OBSERVATIONS:

(1) Diameter of the Tube, $d =$ mm.

(2) Length of tube $L = 100\text{cms}$

VARIABLE OBSERVATIONS:

Sr. No	H_1	H_2	$H =$ $(H_1 + H_2)/2$	Volume of oil collected in Measuring Cylinder, ml	Time t sec. for collection of volume.

CALCULATIONS:

(1) Discharge $Q = \text{Volume} / \text{Time}$

$$(2) \Delta p = (p_1 - p_2) = \rho g h$$

$$(3) \mu = \frac{\pi d^4 (p_1 - p_2)}{128 Q L}$$

GRAPH: Draw the graph between h v/s Δp

CONCLUSION:

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EXPERIMENT NO. 3

SEEPAGE ANALYSIS THROUGH AN EARTH DAM BY USING VISCOUS ANALOGY

AIM :

To determine the hydraulic conductivity of an earth dam using viscous flow analogy.

APPARATUS:

- (1) Dam section consists of two Perspex sheets kept closely parallel (1mm) (vertical model)
- (2) Measuring Jar
- (3) Timer
- (4) Oil of known viscosity

THEORY:

Analog means similarity. Analog models can be applicable to ground water flow by considering two dimensional flow. Generally two types of analog models are used for ground water problems namely,

(i) mechanical analogy and (ii) electrical analogy.

To simulate the ground water flow in laboratory viscous fluid is used and allow to pass through the closely fixed parallel plates. Viscous fluid such as leen seed oil or glycerin is used to determine the aquifer property or of an earth dam material. The loss between two closely plates is given by

$$h_l = \frac{12 \mu V l}{\rho g B^2} . \text{ By taking } B = 2a$$

where '2a' is the spacing between the plates,

the equation can be written as

$$V_m = \frac{a^2 g h_l}{3\nu l} = k_m \frac{h_l}{l}$$

where $\frac{h_l}{l}$ is the gradient and k_m is model coefficient of permeability.

This will follow the Darcy's law. The Darcy's law is $V_p = k i$

where V_p is prototype velocity.

The velocity ratio of model to prototype is given as

$$V_r = \frac{V_m}{V_p} = \frac{a^2 g}{3\nu k}.$$

The field permeability can be given as $K = \frac{V}{\mu}$. Here K is intrinsic permeability depending upon soil characteristics.

For the given boundary conditions flow lines and equipotential lines can be constructed orthogonally so that flow net could be obtained in the form of curvilinear squares. The potential drop between two equipotential lines is Δh and will be same between any two equipotential lines. If H is the total seepage head and N_d is number of potential drops then

$$\Delta h = \frac{H}{N_d}.$$

The discharge between any two stream lines is same is given as

$$\Delta q = k \frac{H}{N_d}.$$

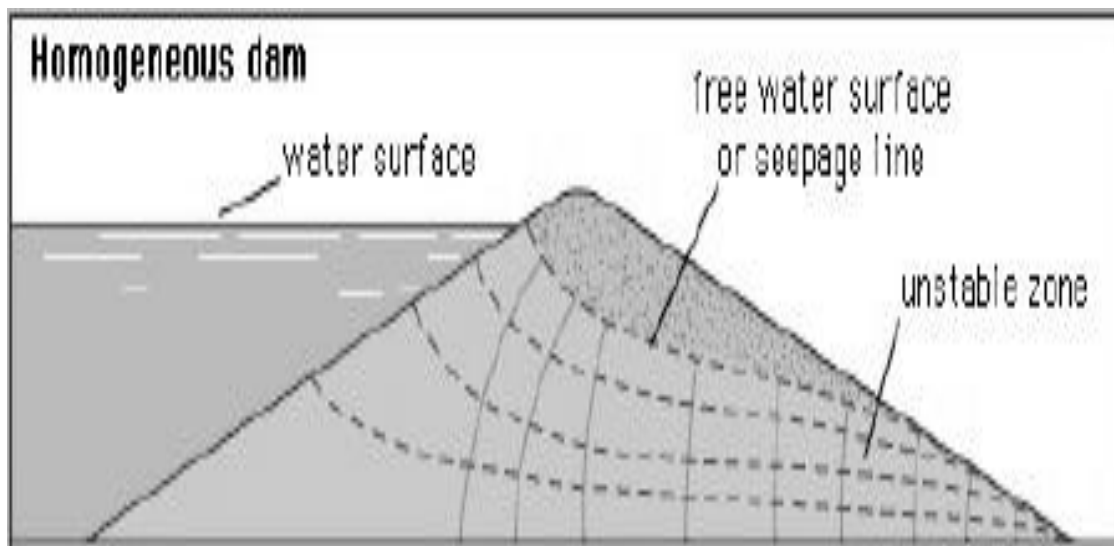
If the N_f is the number of spaces between two flow lines then the seepage discharge will be $q = k H \frac{N_f}{N_d}$

PRACTICAL SIGNIFICANCE:

The plate spacing and viscous fluid is so selected that it will represent desired value of permeability. The water table or piezometric heads can be simulated in vertical models. The flow line across the earthen dam can be witnessed if dye is added. The parallel model is isotropic with respect to permeability.

PROCEDURE:

- 1 Obtain the steady flow from the u/s side (reservoir).
- 2 Note down the coordinates of the seepage line.
- 3 Measure the flow rate of the oil.
- 4 Repeat 'a' to 'c' steps by increasing the head in the reservoir.



FIXED OBSERVATIONS:

Space between the plates $B =$

Upstream angle $\alpha =$

D/s face angle $\beta =$

Height of dam, cms $=$

Viscosity of oil $\mu =$

VARIABLE OBSERVATIONS:**Set -1** Head H_1 , cm =Length L_1 =Distance d_1 =

Volume of oil collected in 60 sec, cc =

Ordinate of seepage line

x																			
y																			

Set -2 Head H_2 , cm =Length L_2 =Distance d_2 =

Volume of oil collected in 60 sec, cc =

Ordinate of seepage line

x																			
y																			

Set -3 Head H_3 , cm =Length L_3 =Distance d_3 =

Volume of oil collected in 60 sec, cc =

Ordinate of seepage line

x																			
y																			

SAMPLE CALCULATION: ()

(1) Flow rate $Q = \frac{\text{Volume of foil}}{\text{Time}}$

(2) Discharge Intensity $q = \frac{Q}{B}$

(3) Distance d_1 $d_1 = d + 0.3L$

(4) Length of discharge face $a = \sqrt{d_i^2 + h^2} - \sqrt{d_1^2 - h^2 \cot^2 \alpha}$

(5) Coeff. Of permeability - analytical $k = \frac{q}{a \sin^2 \alpha}$

(6) Coeff. Of permeability – from flow net $k = \frac{q}{h \frac{N_f}{N_d}}$

COMPUTATIONAL TABLE :

Sr. No.	Head , h cm	Discharge ,q m ³ /s/m	Distance ,d	Length of discharge face, cm	Coefficient of permeability	
					K, analytical	K,from flow net
1						
2						
3						

GRAPH: Draw the flow net fro coordinates of seepage line.

CONCLUSION:

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EXPERIMENT NO. 4 TO STUDY COMPRESSIBLE FLOW

AIM:

To study familiarize with the phenomena and equation of compressible flow.

Objectives:

Here the main objective is the Engineering Equation Solver (EES) . Compressible Flow analysis is having with scores of complicated algebraic equations because of law of thermodynamics are also included. The equation of continuity , momentum along with the equation of state are also needed for analysis of the flow.

Introduction:

The compressibility of a fluid is, basically, a measure of the change in density that will be produced in the fluid by a specific change in pressure and temperature. In general, gases are highly compressible and liquids have a very low compressibility. Many of the compressible flows that occur in engineering practice can be adequately modeled as a flow through a duct or stream tube whose cross-sectional area is changing relatively slowly in the flow direction.

Applications of compressible flow:

- | | |
|-----------------------------|--------------------------------------|
| (1) Aircraft design | (2) Gas and steam turbines |
| (3) Reciprocating engines | (4) Natural Gas Transmission lines |
| (5) Combustion chambers | |

Fundamental Assumptions:

1. The gas is continuous.
2. The gas is perfect (obeys the perfect gas law).
3. Gravitational effects on the flow field are negligible.
4. Magnetic and electrical effects are negligible.

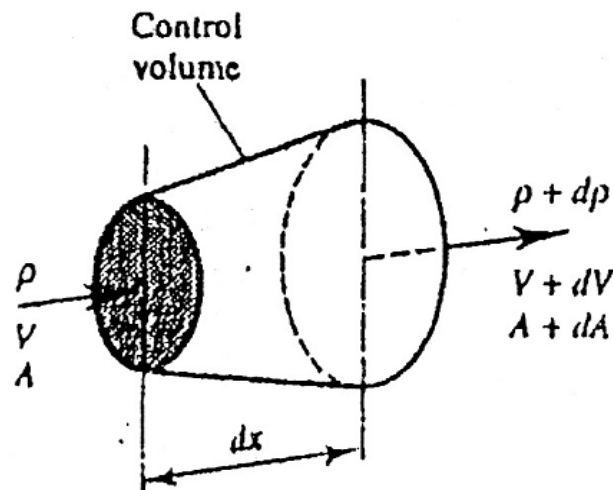
5. The effects of viscosity are negligible.

Applied principles:

1. Conservation of mass (continuity equation)
2. Conservation of momentum (Newton's law)
3. Conservation of energy (first law of thermodynamics)
4. Equation of state

Continuity Equation:

The continuity equation is obtained by applying the principle of conservation of mass to flow through a control volume. One-dimensional flow is being considered. There is no mass transfer across the control volume. The only mass transfer occurs through the ends of the control volume.



Mass enters through the left hand face of the control volume be equal to the rate at which mass leaves through the right hand face of the control volume.

$$\dot{m}_1 = \dot{m}_2$$

$$\rho_1 V_1 A_1 = \rho_2 V_2 A_2$$

In differential form,

$$\frac{d\rho}{\rho} + \frac{dV}{V} + \frac{dA}{A} = 0$$

Bernoulli's Equation Or Energy Equation:

(a) Isothermal Process:

$$\frac{p}{\rho g} \ln(p) + \frac{V^2}{2g} + z = \text{const.}$$

(b) Adiabatic Process:

$$\left(\frac{k}{k-1} \right) \frac{p}{\rho g} + \frac{V^2}{2g} + z = \text{const.}$$

Mach number:

It is the ratio of square root of inertia force to elastic force. It is denoted by M. It is dimension less number. It is also defined as ratio of velocity of fluid or body moving in fluid to the velocity of sound. Mach number provides information that weather the flow is subsonic flow, sonic flow or supersonic flow.

$$\text{Sonic Velocity in terms of Bulk Modulus } C = \sqrt{\frac{K}{\rho}}$$

$$\text{Sonic Velocity in isothermal process } C = \sqrt{\frac{p}{\rho}} = \sqrt{RT}$$

$$\text{Sonic velocity in Adiabatic Process } C = \sqrt{k RT}$$

Mach Number

$$M = \frac{V}{C} = \frac{\text{Vel. at pt. in a fluid}}{\text{Vel. of sound at that pt. at given instant of time}}$$

Subsonic flow ($M < 1$):

A flow is subsonic flow if the Mach number is less than 1. In subsonic flow, velocity of flow (V) is less than velocity of sound wave (C). ($V < C$).

Sonic flow ($M = 1$):

A flow is sonic flow if the Mach number is equal to 1. In sonic flow, velocity of flow (V) is equal to velocity of sound wave (C). ($V = C$).

Supersonic flow ($M > 1$):

A flow is supersonic flow if the Mach number is greater than 1. In sonic flow, velocity of flow (V) is greater than velocity of sound wave (C). ($V > C$).

1. A gas is flowing through a horizontal pipe of cross sectional area of 30 cm^2 . At a point the pressure is 30 N/cm^2 (gauge) and temperature 20°C . At another section the area of cross section is 15 cm^2 and pressure is 15 N/cm^2 (gauge). If the mass flow rate is 0.15 kg/sec , find the velocities of the gas at these two sections, assuming an isothermal change. Take, $R = 287 \text{ J/kg K}$, an atmospheric pressure $= 10 \text{ N/cm}^2$.

2. A gas velocity of 350 m/sec is flowing through a horizontal pipe at a section where pressure is 8 N/cm^2 (absolute) and temperature 30°C . The pipe changes in diameter and at this section the pressure is 12 N/cm^2 (absolute). Find velocity of gas at this section if flow is adiabatic. Take, $R = 287 \text{ J/kg K}$ and $\gamma = 1.4$.

3. Calculate Mach number at a point on jet propelled air craft which is flying at 900 km/hour where temperature is 15 °C. Take, $R = 287 \text{ J/kg K}$ and $\gamma = 1.4$.

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EXPERIMENT NO. 5

DETERMINATION OF COMPRESSIBILITY CORRECTION FACTOR (CCF)

Objective: - To determine the compressibility Correction Factor For Compressible (CCF) for Fluid Flow.

Apparatus: - Pitot Static Tube with coefficient of discharge value 1.00

Induced Flow Wind Tunnel

Anemometer for measurement of speed (RPM)

Differential Manometer connected to pitot tube.

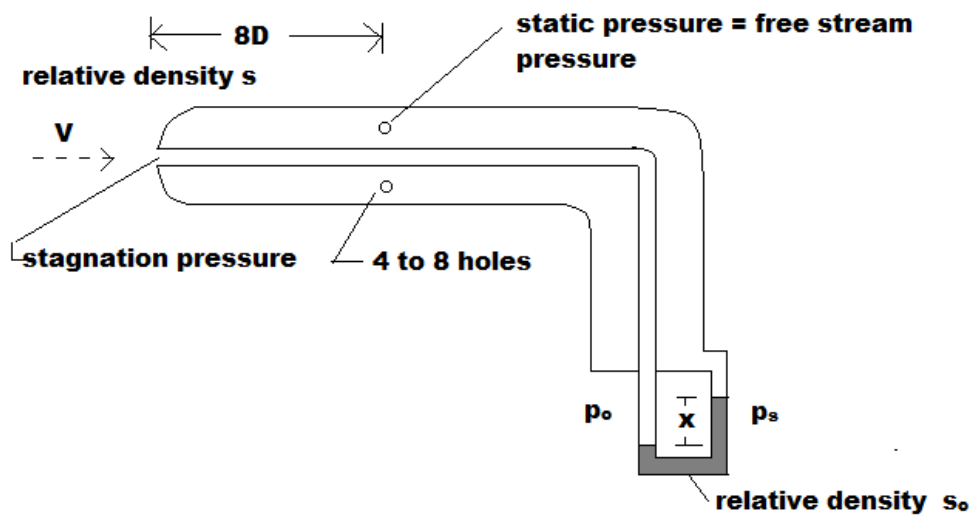
Theory:- In case of pipe flow , Pitot tube is simple device for obtaining velocity by measuring stagnation pressure at the tip of tube and static pressure. The stagnation pressure p_o for a constant density fluid is $p_o = p_s + \rho \frac{V^2}{2}$. For compressible fluid flow the effect of compressibility is isolated in the bracket and that effect solely depends upon the Mach number. The equation is as under

$$p_o = p_s + \rho \frac{V^2}{2} \left[1 + \frac{M^2}{4} + \frac{2-k}{24} M^4 + \dots \right]$$
 It observed experimentally that if $M < 0.2$ The compressibility affects the pressure difference ($p_o - p_s$) is less than 1% the effect of compressibility is not considered and constant density formula is sufficiently accurate. The quantity in the bracket is termed as Compressibility Correction Factor (CCF) because for high value of M the compressibility must be considered. The aircraft speed needs to be corrected to take in to account the effect of compressibility.

The Mach Number $M = V/c$ where c is sound velocity. In case of the compressible fluid flow the pressure and temperature changes due to passage of sound wave is very small but makes the process adiabatic. From the perfect gas law $p = \rho RT$ shows that the speed of sound is the function of the absolute temperature only.

Procedure:-

- In the test section of wind tunnel (which can be considered as the closed conduit) keep the Pitot tube probe facing the flow of air and the the tapings (both Stagnation and static pressure) should be connected to differential (air – water) manometer.
- Let allow the air flow passes through the wind tunnel by starting the blower provided at the d/s end of the tunnel.
- Measure the difference of both limbs of manometer and obtain the difference and compute the head in terms of air.
- Find the local velocity (of flowing air) V .
- Note down the room temperature and convert in to absolute one. Find the ρ at the room temperature from table. Take the ‘R’ Gas constant for air.
- Find sound velocity and Mach number M .
- Insert the value of M in Bracketed terms and find the CCF and find correct velocity.



Fixed Observations:- (1) Room Temperature $C^{\circ} =$

(2) Density of air $\rho \text{ Kg/M}^3 =$

(3) Gas Constant $R = 287 \text{ N-m/kg-K}$ for air.

(4) Density of water, $\rho_w = 1000 \text{ kg/m}^3$

Variable observations:-

Sr. No.	Speed in RPM	$X = \Delta h$ from manometer (difference of two limbs)

Sample Calculation :- (Observation NO. ____)

(1) Head in terms of Air: $H = \frac{\rho_w H_w}{\rho_{air}}$

(2) Local velocity $V = \sqrt{2gH_{air}}$

(3) Sound velocity $C = \sqrt{kRT}$

(4) Mach Number $M = \frac{V}{C}$

(5) Compressibility Correction Factor (CCF) = $\left[1 + \frac{M^2}{4} + \frac{2-k}{24} M^4 + \right]$

(6) Corrected Velocity = CCF x V

Computational Table:

Sr. No.	Head In terms of Air H_{air} , m	$V = \sqrt{2gH_{air}}$ m/s	Mach No. $M=V/C$	Comp.Correction Factor -CCF	Corrected velocity $V = CCF \times V$
1					
2					
3					
4					
5					
6					

Conclusions:-

Questions:- (a) How do you correct velocity of aircraft when Mach Numbers very high?

(b) Collect some useful numbers for air.

(c) What will occur when $M > 1$ changes to $M < 1$?

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EXPERIMENT NO 6

2D VELOCITY PROFILE OF A TURBULANT FLOW IN CLOSED CONDUITS

OBJECTIVE:

To determine 2D velocity profile of a turbulent flow in closed conduits

Apparatus: - Pitot Static Tube with coefficient of discharge value 1.00

Induced Flow Wind Tunnel

Anemometer for measurement of speed (RPM)

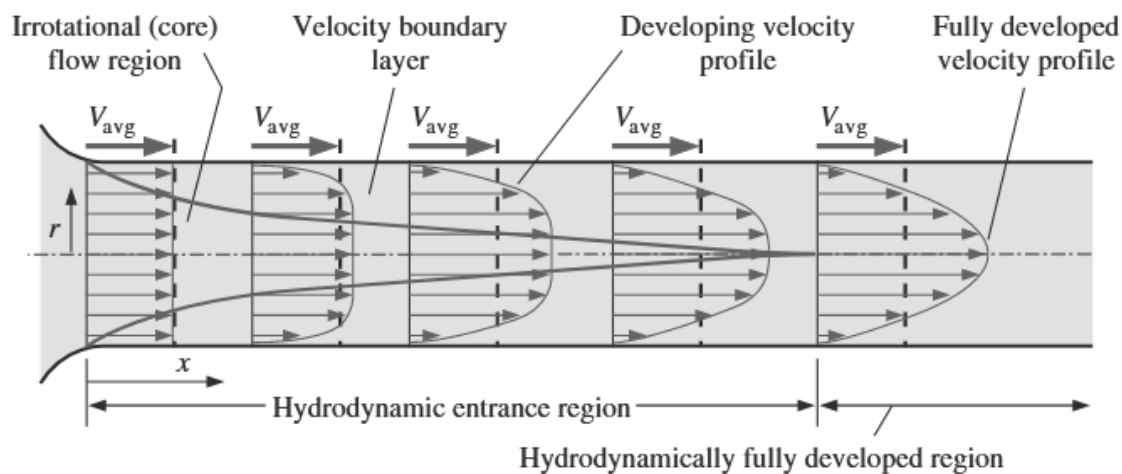
Differential Manometer connected to pitot tube.

Theory:-

Consider a fluid entering a circular pipe at a uniform velocity. Because of the no-slip condition, the fluid particles in the layer in contact with the surface of the pipe come to a complete stop. This layer also causes the fluid particles in the adjacent layers to slow down gradually as a result of friction. To make up for this velocity reduction, the velocity of the fluid at the mid-section of the pipe has to increase to keep the mass flow rate through the pipe constant. As a result, a velocity gradient develops along the pipe.

The region of the flow in which the effects of the viscous shearing forces caused by fluid viscosity are felt is called the boundary layer. The hypothetical boundary surface divides the flow in a pipe into two regions: the boundary layer region, in which the viscous effects and the velocity changes are significant, and the irrotational(core) flow region, in which the frictional effects are negligible and the velocity remains essentially constant in the radial direction. The thickness of this boundary layer increases in the flow direction until the boundary layer reaches

the pipe centre and thus fills the entire pipe, as shown in Figure below. The region from the pipe inlet to the point at which the boundary layer merges at the centre line is called the hydrodynamic entrance region, and the length of this region is called the hydrodynamic entrance length L_h . Flow in the entrance region is called hydrodynamically developing flow since this is the region where the velocity profile develops. The region beyond the entrance region in which the velocity profile is fully developed and remains unchanged is called the hydrodynamically fully developed region. The flow is said to be fully developed, when the normal sized temperature profile remains unchanged as well.



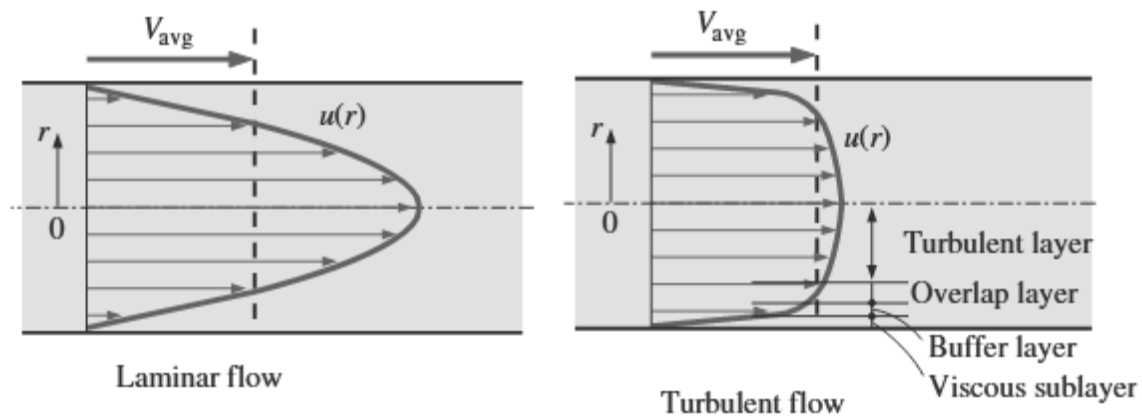
Development of the velocity boundary layer in a pipe

(The developed average velocity profile is parabolic in laminar flow, but somewhat flatter or fuller in turbulent flow)

In fluid flow, it is convenient to work with an average velocity V_{avg} , which remains constant in incompressible flow when the cross-sectional area of the pipe is constant. Typical velocity profiles for fully developed laminar and turbulent flows are given in figure below. Note that the velocity profile is parabolic in laminar flow but is much fuller in turbulent flow, with a sharp drop near the pipe wall.

Turbulent flow along a wall can be considered to consist of four regions, characterized by the distance from the wall.

1. The very thin layer next to the wall where viscous effects are dominant is the viscous (or laminar or linear or wall) sub-layer. The velocity profile in this layers very nearly linear, and the flow is streamlined.
2. Next to the viscous sub-layers the buffer layer, in which turbulent effects are becoming significant, but the flow is still dominated by viscous effects.
3. Above the buffer layer is the overlap (or transition) layer, also called the inertial sub-layer, in which the turbulent effects are much more significant, but still not dominant.
4. Above that is the outer (or turbulent) layer in the remaining part of the flow in which turbulent effects dominate over molecular diffusion (viscous) effects.



The velocity profile with radial distance for laminar and turbulent flow in a pipe

Fixed Observations:- (1) Room Temperature $C^0 =$

(2) Density of air ρ $\text{Kg/M}^3 =$

(3) Gas Constant $R = 287 \text{ N-m/kg-K}$ for air.

(4) Density of water, $\rho_w = 1000 \text{ kg/m}^3$

(5) Speed in RPM =

Variable observations:-

Sr. No.	Distance from boundary (Top)	X = Δh from manometer (difference of two limbs)		
		H1	H2	Δh
1				
2				
3				
4				
5				
6				
7				
8				
9				

Sample Calculation :- (Observation NO. ____)

(1) Head in terms of Air: $H = \frac{\rho_w H_w}{\rho_{air}}$

(2) Local velocity $V = \sqrt{2gH_{air}}$

(3) Sound velocity $C = \sqrt{kRT}$

(4) Mach Number $M = \frac{V}{C}$

(5) Compressibility Correction Factor (CCF) = $\left[1 + \frac{M^2}{4} + \frac{2-k}{24} M^4 + \right]$

(6) Corrected Velocity = CCF x V

(7) Reynold's No. $Re = \rho V D / \mu$

Computational Table:

Sr. No.	Head In terms of Air H_{air} , m	$V = \sqrt{2gH_{air}}$ m/s	Mach No. $M = V/C$	Comp.Correction Factor -CCF	Corrected velocity $V = CCF \times V$	Re
1						
2						
3						
4						
5						
6						
7						
8						
9						

Graph :- Draw the 2D velocity profile with variation of radial distance for the flow.

Conclusions:-

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EXPERIMENT NO 7

FLOW IN CLOSED CONDUITS

OBJECTIVE:

To determine co-efficient of friction in closed conduits.

APPARATUS:

- Pipes of different diameter
- Measuring Tank.
- Stop watch.
- U-Tube Manometer.

THEORY:

For laminar flow , head loss, h_f varies first power of velocity while in turbulent flow the h_f varies to the second power of velocity. Head loss in closed conduit also depends on the roughness of the conduit material.

German engineer Nikuradse carried out several pipe flow experiments in which pipes inner surface was coated with sand grain. By dimensional analysis it is shown that the friction factor 'f' is a function of Reynolds number and relative roughness $\frac{\epsilon}{d}$. In case of the artificial roughness , roughness is uniform where as in commercial pipes it is irregular both in size and distribution. However the roughness of commercial pipes may be described by ϵ , which means that the pipe has same value of 'f' at higher Reynolds numbers would be obtained if smooth pipe was coated with sand grain of uniform size ϵ . Blasius shown that Reynolds Number between 3000 and 100,000 the friction factor maybe expressed as

$$f = 0.3164 / R^{0.25} .$$

For pipe flow , the viscous sub layer completely submerges the height of roughness element and friction factor 'f' is written as

$$\frac{1}{\sqrt{f}} = 2 \log R_e \sqrt{f} - 0.8$$

at very high Reynolds number the pipe will be hydraulically rough

$$\frac{1}{\sqrt{f}} = 2 \log \frac{d}{\epsilon} + 1.14 .$$

For the transitional stage Colebrook formula is available.

Practical Significance:-

In foregoing discussion it is very important to use the formula for ‘f’. So it is preferable to obtain numerical values from Moody’s diagram. Value $\frac{\epsilon}{d}$ for commercial pipes may conveniently be found from Moody’s diagram. The value of ‘f’ may have error of $\pm 5\%$ for smooth pipes and of $\pm 10\%$ for rough pipes. For old pipes the value of ϵ may be much higher but much depends on the type of the fluid being transported. “f” generally have the value between 0.0 to 0.07.

TECHNICAL DATA:

Total Length: 1500 mm.

Test Length: 1000 mm.

Pipe 1: OD – 21 mm, ID – 17 mm.

Pipe 2: OD – 15 mm, ID – 12 mm.

Pipe 3: OD – 12 mm, ID – 10 mm.

PROCEDURE:

1. Fill the sump tank with water.
2. Attach the limbs of the manometer with the help of the flexible pipe to the pressure tapping of the pipe across which the pressure drop is to be determined.
3. Switch on the pump and adjust the flow through the pipe with the help of supply valve and bypass valve provided.
4. Remove any air from the system by opening the valve provided on the manometer.
5. After steady state is achieved record the difference in manometer limbs.
6. Collect about known quantity (height difference) of water in the collection tank with liquid level indicator and note down the time required for the same.

7. Repeat the above at least four-five times starting from the minimum flow rate and going to the maximum value.
8. Select another pipe and repeat the experiment.

OBSERVATION:

Fixed Observation:-

Pipe – 1 OD = 21 mm ID = 17 mm

Pipe – 2 OD = 15 mm ID = 15 mm

Pipe – 3 OD = 12 mm ID = 10 mm

Testing Length of Pipes L = 1.00 m Kinematic Viscosity of water ν = m²/s

Variable observations:-

Pipe No: 1 OD = 21 mm ID = 17 mm	Sr.No.	Time for collecting 1000 cc of water,s	Loss of head across the pipe ,cm f water
	1		
	2		
	3		
	4		
	5		
	6		
Pipe No: 2 OD = 15 mm ID = 13 mm	Sr.No.	Time for collecting 1000 cc of water,s	Loss of head across the pipe ,cm of water
	1		
	2		
	3		
	4		
	5		
	6		

Pipe No: 3 OD = 12 mm ID = 10 mm	Sr.No.	Time for collecting 1000 cc of water,s	Loss of head across the pipe ,cm of water
	1		
	2		
	3		
	4		
	5		
	6		

SAMPLE CALCULATION:

(Pipe No: 1 Observation No:- _____)

$$(1) \text{ Velocity } V = \frac{\text{Volume of water collected}}{\text{Time for collection} \times c/s \text{ area of pipe}} =$$

=

$$(2) \text{ Reynolds Number } R = \frac{VD_1}{\nu} =$$

=

$$(3) \text{ Friction Factor } f =$$

=

(Pipe No: 2 Observation No:- _____)

$$(1) \text{ Velocity } V = \frac{\text{Volume of water collected}}{\text{Time for collection} \times c/s \text{ area of pipe}} =$$

=

(2) Reynolds Number $R = \frac{VD_1}{\nu}$

=

=

(3) Friction Factor f

=

=

(Pipe No: 3 Observation No:- _____)

(1) Velocity $V = \frac{\text{Volume of water collected}}{\text{Time for collection} \times c/s \text{ area of pipe}}$

=

=

(2) Reynolds Number $R = \frac{VD_1}{\nu}$

=

=

(3) Friction Factor f

=

=

Computation Table:-

(Pipe No: 1)

Sr.No.	Log $\frac{h_f}{l}$	Velocity V ,cm/s	Log V	Reynolds NO: R	Log R	Friction factor f	Log f

(Pipe No: 2)

Sr.No.	Log $\frac{h_f}{l}$	Velocity V ,cm/s	Log V	Reynolds NO: R	Log R	Friction factor f	Log f

(Pipe No: 3)

Sr.No.	$\text{Log } \frac{h_f}{l}$	Velocity V ,cm/s	Log V	Reynolds NO: R	Log R	Friction factor f	Log f

GRAPHS:- (1) draw a graph of $\log \frac{h_f}{l}$ v/s $\log V$

(2) draw a graph of $\log R$ v/s $\log f$

CONCLUSION:

Marks obtained:	Signature of faculty:	Date:
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Date:

Fluid Mechanics-2 (CL372)

EXPERIMENT NO. 8

TO STUDY BOUNDARY LAYER FLOW

AIM:

To understand the concept of boundary layer flow characteristics

OBJECTIVES:

When body immersed in a fluid stream the viscous effect near the body surfaces and its wake region are the area of study at different Reynolds number. The flow stream adjacent to the immersed body is unconfined boundary layer flow.

Introduction to boundary layer:

External flows around streamlined bodies at high Reynolds number have viscous (shear and no-slip) effects confined close to the body surfaces and its wake, but are nearly in-viscid far from the body.

Ideal fluid theory assumes that fluid is ideal, zero viscosity and constant density. Results obtained don't match with experiments. With ideal fluid, there is no drag force. However, in practice, drag force exists. In practice, fluids adhere to the boundary (No slip condition). For fluids having low viscosity the effect of internal friction in a fluid is appreciable only in a narrow region surrounding the fluid boundaries. This will setup steep velocity gradient on the boundaries. This variation in velocity near the wall gives rise to shear stresses. Outside the narrow region the flow may be considered ideal or potential flow. Boundary layer is a very thin layer in which the effect of viscosity is felt even at high Reynolds number. Of course this layer is in the immediate vicinity of the wall (or boundary).

Applications of boundary layer:

- Aerodynamics (airplanes, rockets, projectiles)
- Hydrodynamics (ships, submarines, torpedoes)
- Transportation (automobiles, trucks, cycles)
- Wind engineering (buildings, bridges, water towers)
- Ocean engineering (buoys, breakwaters, cables)

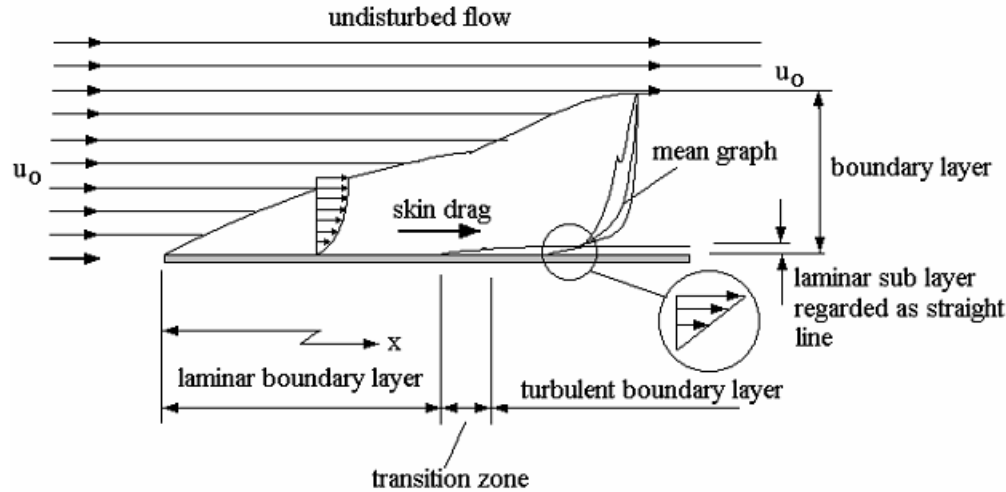
Boundary layer concept:

When a real fluid flows past a solid boundary, there develops a thin layer very close to the boundary in which the velocity rapidly increases from zero at the boundary (due to no slip condition) to the nearly uniform velocity in the free stream. That fluid layer which has had its velocity affected by the boundary shear is called the boundary layer.

The boundary layer is very thin at the leading edge of the plate but continual action of the shear stress to slow down additional fluid particles causing the thickness of boundary layer to increase with distance from the upstream point. The fluid in the layer is subjected to the pressure gradient and that will increase the momentum if the pressure gradient decreases in the direction of the flow.

In the beginning the boundary layer starts as a Laminar B.L. in which the fluid particle moves in smooth layer. As the laminar boundary layer increases in its thickness, it becomes unstable and ultimately it transforms into turbulent boundary layer. When the boundary layer has become turbulent, there is still a very thin layer next to the boundary that has a laminar motion. It is called laminar sublayer.

Consider a steady, uniform stream of fluid moving with velocity U_o on a flat plate. At the leading edge, the thickness of the boundary layer is zero. In the downstream direction, the thickness of the boundary layer goes on increasing as shown in figure.



Various definitions of boundary layer thicknesses δ have been suggested.

Boundary layer thickness (δ): It is the distance from the boundary in which the local velocity reaches 99% of the main stream velocity and is denoted by (δ). $y = \delta, u = 0.99 U_o$

Displacement thickness (δ_i): It is defined as the distance perpendicular to the solid boundary by which the main flow will have to be displaced outward so that the actual discharge would be same as that of the ideal fluid past the displaced boundary.

$$\delta_i = \int_0^{\delta} \left(1 - \frac{u}{U_o} \right) dy$$

Momentum thickness (θ): It is defined as the distance measured perpendicular from the actual boundary such that the momentum flux through this distance is equal to the deficit of the momentum flux due to the boundary layer formation. It is denoted by θ .

$$\theta = \int_0^{\delta} \frac{u}{U_o} \left(1 - \frac{u}{U_o} \right) dy$$

Energy Thickness (δ^{}):** It is the distance perpendicular to the boundary by which the boundary has to be displaced to compensate for the reduction in the kinetic energy of the fluid caused due to the formation of the boundary layer. It is denoted by δ^{**}

$$\delta^{**} = \int_0^{\delta} \frac{u}{U_o} \left(1 - \frac{u^2}{U_o^2} \right) dy$$

In case of the turbulent boundary layer the velocity is more uniform and the thickness is given as

$$\delta = \frac{0.372 \cdot x}{(R_x)^{1/5}} \quad \text{when} \quad 5 \times 10^5 < R_e < 10^7 \quad \text{for power law velocity distribution.}$$

The laminar sub-layer, although very thin is an important factor in deciding whether a surface is hydro-dynamically smooth or rough surface.

Shape Factor (H): It is defined as the ratio of the displacement thickness to the momentum thickness. $H = (\delta^*/\theta)$

QUESTIONS:

1. Give some practical examples of boundary layer that you come across everyday life.
2. Is the flow within the boundary layer rotational or irrotational?
3. Why boundary layer thickness increases as flow past the flat plate leading edge?
4. What is laminar sublayer? How is the concept of laminar sublayer useful?

Marks obtained:	Signature of faculty:	Date:
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EXPERIMENT NO. 9

PRESSURE DISTRIBUTION AROUND CYLINDER

OBJECTIVE:

To study the pressure distribution around the cylinder of infinitive length (two dimensional) in wind tunnel and comparing with the potential flow pressure distribution.

APPARATUS:

- Wind tunnel with induced / forced blower (fan),
- Wooden cylinder whose length is equal to width of test section,
- Pitot tube
- Manometer.

THEORY:

Pressure distribution for ideal fluid flow over the surface of a circular cylinder is given by

$$\frac{p - p_o}{\frac{1}{2} \rho U_o^2} = 1 - 4 \sin^2 \theta$$

in which p_o and U_o are the pressure and free stream velocity of undisturbed flow θ is measured anticlockwise w.r.t d/s stagnation point. For ideal fluids the pressure distribution is symmetrical about the flow direction no lift or no drag developed. This is described as under.

$$\frac{p - p_o}{\frac{1}{2} \rho U_o^2} = 1 - 4 \sin^2 \theta \quad \text{at } \theta = 0 \text{ and } \pi$$

$$\frac{p - p_o}{\frac{1}{2} \rho U_o^2} = 1 - 4 \sin^2 \theta \quad \text{at } \theta = \pi/2 \text{ and } 3\pi/2$$

$$\frac{p - P_o}{\frac{1}{2} \rho U_o^2} = 1 - 4 \sin^2 \theta \quad \text{at } \theta = \pi/6 \text{ and } 5\pi/6.$$

But the for real fluid the pressure distribution profile is different where the boundary layer separates from surface of cylinder. That can be determined with experiment.

Practical Significance It has got many application like pressure or heat transfer rate from the bundles in heat exchange equipments. Design of cooling towers and chimneys require a pressure estimation on leeward side i.e. wake regions.

PROCEDURE:

- (1) Start the blower and let the air flow becomes steady.
- (2) Measure u/s total pressure, by keeping the probe of pitot tube facing the air velocity.
- (3) Measure the static pressure p_{st} of free stream velocity
- (4) On the surface of the cylinder at predetermined angle (increasing the clockwise direction) take the pressure p_{sc} . Compute C_p and C_{act} for all points.

OBSERVATION:

(1) Free stream total pressure p_{to} =

(2) Free stream static pressure p_{st} =

(3) Dynamic pressure $p_{to} - p_{st} = \frac{1}{2} \rho U^2$ =

Variable Observations:-

Sr. No.	Angular position on surface of cylinder	Static pressure at each point $P_{sc}, \text{cm of water}$			Theoretical pressure coefficients C_p	Actual pressure coefficient, C_{act}
		L L	RL	Diff		

SAMPLE CALCULATION:

(1) static pressure on the surface of the cylinder p_{sc} =

(2) Theoretical pressure coefficient $C_p = 1 - 4 \sin^2 \theta$ =

(3) Actual pressure coefficient $C_{pact} = \frac{p - p_o}{\frac{1}{2} \rho U_o^2}$ =

GRAPHS:-

1. Draw a graph θ v/s C_{pt} and
2. θ v/s C_{act}

CONCLUSION:

Marks obtained:	Signature of faculty:	Date:
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EXPERIMENT NO. 10

TO DETERMINE THE LIFT FORCE AND LIFT COEFFICIENT OF AIR FOIL

OBJECTIVE:

To determine the lift force and lift coefficient of symmetric air foil by measurement of pressure on upper and lower surface of it.

APPARATUS:

- Wind tunnel – subsonic,
- NACA 0015 aero foil ,
- manometer and pitot tube

TERMINOLOGY:

Aerofoil : It is defined as the streamline body designed for minimum drag and more lift. Chord Line: is a straight line joining centers of curvature of leading edge and trailing edge.

Chord length(C): the length of chord line between leading edge and trailing edge

Camber line: the center line of aerofoil section.

Camber (δ) : is the maximum distance between the camber line and chord line.

Percentage camber : $(100 \delta / C)$

Span(B): the aerofoil length in the direction perpendicular to the cross section.

Plan Area (A) :- $C \times B$

Mean chord length : $C^- = A/B$

Aspect Ratio: $AR = (\text{Span})/(\text{mean Chord}) = B/C^- = B^2/A$

Deviation θ = angle between tangent to the camber line at the trailing edge and tangent to the camber at the leading edge.

Angle of Attack: the angle between the direction of relative motion and chord line.

Pressure Coefficient: $C_p = \frac{p_{sc} - p_o}{\frac{1}{2} \rho U_o^2}$ where p_{sc} is local pressure and p_o is the pressure far upstream of

aerofoil where velocity is V_o .

National Advisory Committee for Aeronautics (NACA) successfully developed such air foil and tested which is widely used in the field of axial flow of turbo machine. The aerofoil is described as per standard laid by NACA

(1) NACA 0016 – symmetrical foil maximum thickness $t_m = 16\%$ of chord length of length C of air foil

(2) NACA 2412 – Max.Camber = m , 2% of chord length C

Position of maximum camber = $X_m = 0.4$ of chord length C from leading edge

Maximum thickness = $t_m = 12\%$ of chord length

THEORY:

The primary object of the aerofoil is to produce lift when submerged in fluid stream and at the same time the aerofoil experiences drag. Here to maximize the lift and minimize drag. When the irrotational flow (rectilinear flow) there will be two stagnation points S_1 and S_2 as shown in Fig 1 . The other type of flow is ideal fluid flow with constant strength of circulation. When combination of rectilinear and circular flow (say clockwise) there will be change in velocity on the upper surface and lower surface of the air foil. The velocity is additive on the upper side and subtractive on the lower side. When the Bernoulli's equation is considered the pressure on the upper side will be lower than the lower side of the air foil. The circulation around the air foil is given as

$$\Gamma = \pi C U_o \sin \alpha$$

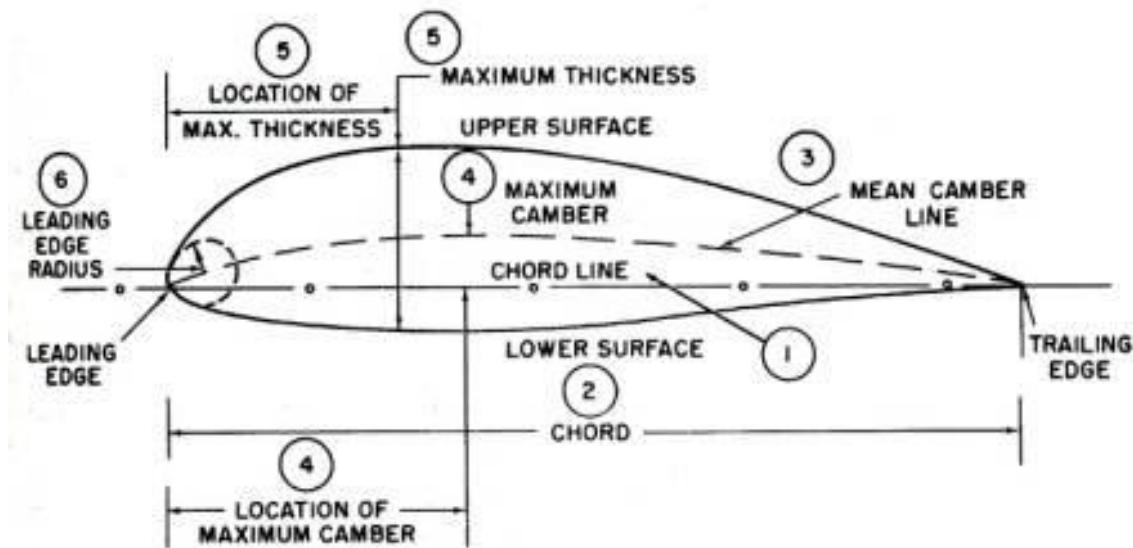
where α is angle of attack from the zero lift direction of air foil. The lift coefficient C_L can be written as

$$C_L = \frac{F_L}{\frac{1}{2} \rho U_o^2 A} = 2 \pi \alpha$$

Real fluid: Air foil , however, does not rotate and follow the different mechanism for producing and maintaining circulation. If streams of unequal velocities meet , there is discontinuity of velocities at their inner faces that produces vortex. The vortex is cast off soon after beginning of motion. It however gives rise to circulation of starting of vortex. If the velocity at upper surface exceeds the velocity of lower surface, the circulation around the foil will be in clockwise direction. Such velocity difference in accordance with BE produces higher pressure on the lower side. Such pressure distribution gives rise to upward thrust namely lift.

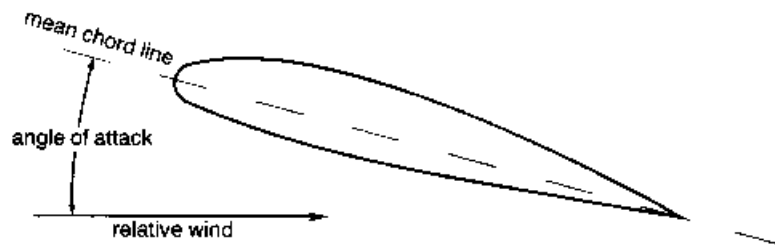
PROCEDURE:

- Symmetric NACA 0015 of chord length 300mm is fitted with small angle of attack in the test section of the wind tunnel. There are holes on upper and lower side of the wind tunnel.
- After starting of the wind tunnel , the total pressure P_o should be measured at far distance on the u/s side of air foil.
- Static pressure p_{sc} is measured at points of the airfoil by pitot tube and differential manometer connection respectively.
- The pressure distribution can be plotted and finally the lift force can be computed.
- Velocities at all points on the surfaces of the air foil can also be found out by using static and total pressure and lift force can be worked out from the circulation around the air

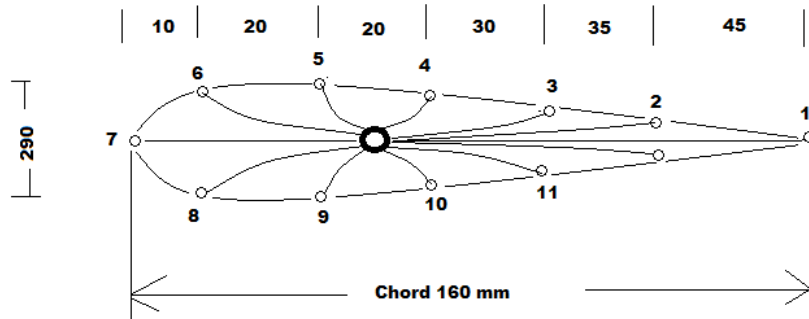


foil.

Image from : www.ace.gatech.edu



NACA 0018



OBSERVATION:

Fixed Observations:-

- (1) Total pressure on u/s side $p_{to} =$
- (2) Static pressure on u/s side $p_{sc} =$
- (3) Dynamic pressure on u/s side $p_{dy} = p_{to} - p_{sc}$
- (4) Density of air $\rho_{air} =$

Variable Observation:-

Sr. NO.	Distance X from Leading edge	Static pressure at point (cm of water)					
		On upper side			On lower side		
		LHS	RHS	Diff.	LHS	RHS	Diff.
1							
2							
3							
4							
5							
6							

SAMPLE CALCULATION:

Sample Calculation:- (Obs. No.)

$$(1) \quad \text{Coefficient of pressure } C_p = \frac{p_{sb} - p_{sc}}{p_{to} - p_{sc}} =$$

Psc = Static pressure =

Pto = Total Pressure =

Psb=Surface Pressure =

$$(2) \text{ Lift Coefficient (from pressure coefficient graph) } C_L = \frac{A}{l_1 C} =$$

$$(3) \text{ Free stream Velocity } U_o = \sqrt{\frac{2 g p_{dy} \times 1000}{\rho_{air}}}$$

$$(4) \text{ Lift force from the graph of pressure distribution: } L = \frac{1}{2} \rho_{air} U_o^2 C_L =$$

$$(5) \text{ Velocity at point on surface } U_{sc} = \sqrt{\frac{2 (p_{to} - p_{sb}) \rho_{water} g}{\rho_{air}}} =$$

(6) Total Pressure at point $p_{to} = p_{sb} + \frac{1}{2} U_o^2 =$

(7) Net area (from vel. Grph) $A_{net} = A_1 - A_2 =$

(8) Circulation Factor $\Gamma = A_{net} \times scale$

(9) Lift coefficient by velocity approach $C_L = \frac{2\Gamma}{l U_o}$

(10) Lift force by velocity distribution: $L = \frac{1}{2} \rho_{air} U_o^2 C_L$

(11) Lift coefficient (Theoretical) $C_L = 2 \Pi \alpha$

(12) Lift force (Theoretical) $L = \frac{1}{2} \rho_{air} U_o^2 C_L$

COMPUTATION TABLE:

Sr. No.	C _p at surface		Velocity V _{sc} at surface	
	Upper	Lower	Upper	Lower
1				

2				
3				
4				
5				
6				

GRAPHS:-

(1) C_p v/s X

(2) V_{sc} v/s X where X is the distance of point from leading edge.

CONCLUSION:

Marks obtained:	Signature of faculty:	Date:
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TILTING FLUME

Experimental Setup:

Hydraulic flume (tilting type) is an equipment for conducting open channel flow experiments. The flume consists of a long channel with transparent sidewalls. A suitable inlet section with baffle screens is provided to obtain a stabilisation zone for the inlet water. A hand operated screw mechanism is provided for tilting the flume to the required slope. A graduated scale fixed exactly 1.0m away from the pivot point on the flume support stand indicates the slope directly.

Two movable gates, one at the inlet and the other at the outlet are provided to control the flow in the test section. A trolley wheel assembly and a gun metal pointer gauge with a pinion and rack arrangement are provided to measure the water level in the test section. Rails are provided on the top of the flume frames for easy movement of the trolley, which carries the hook/pointer gauge.

Water from the supply pump is supplied to the flume inlet section through a gate valve and a flexible hose. This water after passing through the stabilisation zone flows in the transparent test section past the models and returns to the sump tank.

The dimensions of the flume are as follows:

1. Total length - 4.0m,
2. Width - 0.15m
3. Maximum Slope - 1 in 25

Water from the flume is returned to the sump tank through a collecting tank, fitted with a hooper arrangement, gauge glass scale and a drain valve. Time taken for collecting a fixed quantity of water is measured and the flow rate is calculated.

Date:

Fluid Mechanics-2 (CL372)

EXPERIMENT NO. 11

TO DETERMINE THE COEFFICIENT OF DISCHARGE OF THE STANDING WAVE VENTURIFLUME

AIM:

To conduct an experiment on the standing wave venturiflume and determine the coefficient of discharge.

THEORY:

A structure constructed across an open channel restricting its width such that the flow rate could be measured is known as a venturiflume. When the flow rate through such a venturiflume is maximum, a standing wave will occur at its downstream. Such a venturiflume is called a standing wave venturiflume or a modular venturiflume.

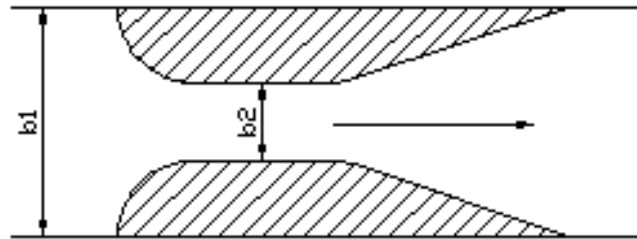
EXPERIMENTAL PROCEDURE:

1. Fix the venturiflume in the flume channel and set the channel slope to be zero. With the pointer gauge, note the position of the bottom surface
2. Open the inlet gate fully and close the outlet gate completely.
3. Allow water in the channel and by slightly opening the outlet gate, create a hydraulic jump on the downstream side of the venturiflume. If the flow is either too much or too little, adjust the supply gate valve correspondingly.

NOTE: The outlet gate has to be opened only about 1-2 cm to achieve this condition.

4. Using the pointer gauge, measure the height of the water level (H meters) in the channel upstream of the venturiflume above the bottom surface.

5. Measure the actual flow rate with the collecting tank.



OBSERVATIONS & CALCULATIONS:

SET 1 :

1. Area of tank a = 0.6×0.5 sq.m

= 0.3 sq.m

Rise of level = _____ m.

Volume collected = _____ cu.m.

Time taken = _____ secs.

Actual discharge Q_a = _____ Cu.m/s

2. Width of channel at inlet $b_1 = 0.15 \text{ m}$

Width of channel at throat $b_2 = 0.05 \text{ m}$

Height of water level = _____ m

Velocity of water at inlet $V_1 = Q/(b_1 H) \text{ m/sec}$

3. Specific energy at inlet $E_{s1} = (V_1^2/2g) + H$

Theoretical discharge $Q_t = 1.71 b_2 (E_{s1})^{1.5} \text{ cu.m/sec}$

Coefficient of discharge $C = Q / Q_t$

SET 2 :

1. Area of tank $a = 0.6 \times 0.5 \text{ sq.m}$

$= 0.3 \text{ sq.m}$

Rise of level = _____ m.

Volume collected = _____ cu.m.

Time taken = _____ secs.

Actual discharge Q_a = _____ Cu.m/s

2. Width of channel at inlet b_1 = 0.15 m

Width of channel at throat b_2 = 0.05 m

Height of water level = _____ m

Velocity of water at inlet V_1 = $Q/(b_1 H)$ m/sec

3. Specific energy at inlet E_{s1} = $(V_1^2/2g) + H$

Theoretical discharge Q_t = $1.71 b_2 (E_{s1})^{1.5}$ cu.m/sec

Coefficient of discharge $C = Q / Q_t$

SET 3 :

1. Area of tank a = 0.6×0.5 sq.m

= 0.3 sq.m

Rise of level = _____ m.

Volume collected = _____ cu.m.

Time taken = _____ secs.

Actual discharge Q_a = _____ Cu.m/s

2. Width of channel at inlet b_1 = 0.15 m

Width of channel at throat b_2 = 0.05 m

Height of water level = _____ m

Velocity of water at inlet V_1 = $Q/(b_1 H)$ m/sec

3. Specific energy at inlet E_{s1} = $(V_1^2/2g) + H$

Theoretical discharge Q_t = $1.71 b_2 (E_{s1})^{1.5}$ cu.m/sec

Coefficient of discharge $C = Q / Q_t$

CONCLUSION:

Marks obtained:	Signature of faculty:	Date:
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Date:

Fluid Mechanics-2 (CL372)

EXPERIMENT NO. 12

TO DETERMINE DISCHARGE OF THE OPEN CHANNEL WITH A SLOPE

AIM:

To conduct an experiment to determine the discharge in an open channel with a slope.

THEORY:

An open channel is a conduit in which water flows with a free surface. The channels are generally built with a small slope such that, water flows under gravity from one point to another. Since the free surface is at atmospheric pressure, the only pressure variation is hydrostatic and the driving force for open channel flows is gravity. The counter acting force on the water flow is the frictional force, which depends upon the area of the wetted surface and the flow velocity. That is, the head available due to the slope of the channel is lost in overcoming the frictional resistance.

EXPERIMENTAL PROCEDURE:

1. Remove all the models in the channel and lift both the upstream and downstream gates such that there is no obstruction to the flow.

2. Raise (or lower) the flume body by rotating the screw rod handle to obtain the required slope. The slope-scale arrangement on flume and support stand directly indicates slope.
3. Using the pointer gauge note the bottom surface level of the channel. Allow water in the channel such that a steady flow is achieved.
4. Using the pointer gauge, measure the water level in the channel. From the earlier measured bottom level, calculate the depth of the water flow.
5. Measure the actual flow rate with the collecting tank.

CALCULATIONS:

SET 1 :

1. Area of tank a = 0.6×0.5 sq.m

= 0.3 sq.m

Rise of level = _____ m.

Volume collected = _____ cu.m.

Time taken = _____ secs.

Actual discharge Q_a = _____ Cu.m/s

2) Slope of the channel = i = _____

Depth of water level = H = _____ m

Width of channel B = 0.15 m

Area of flow a = B H = _____sq.m

Velocity of flow $V = Q/(B H) = \underline{\hspace{2cm}} \text{ m/sec}$

Hydraulic mean radius m = $BH/(B+2H)$ = _____ m

by Chezy's formula,

$$V = C \sqrt{mi}$$

Hence, C, Chezy's constant is given by

$$\underline{C} = \underline{V} / \sqrt{m_i}$$

SET 2 :

1. Area of tank a = 0.6×0.5 sq.m

= 0.3 sq.m

Rise of level = _____m.

Volume collected = _____cu.m.

Time taken = _____secs.

Actual discharge Q_a = _____ Cu.m/s

2) Slope of the channel = $i =$ _____

Depth of water level = $H =$ _____ m

Width of channel $B = 0.15$ m

Area of flow $a = B H =$ _____ sq.m

Velocity of flow $V = Q/(B H) =$ _____ m/sec

Hydraulic mean radius $m = BH/(B+2H) =$ _____ m

by Chezy's formula,

$$V = C \sqrt{mi}$$

Hence, C , Chezy's constant is given by

$$C = \frac{V}{\sqrt{mi}}$$

SET 3 :

1. Area of tank $a = 0.6 \times 0.5 \text{ sq.m}$

$$= 0.3 \text{ sq.m}$$

Rise of level $= \underline{\hspace{2cm}} \text{m.}$

Volume collected $= \underline{\hspace{2cm}} \text{cu.m.}$

Time taken $= \underline{\hspace{2cm}} \text{secs.}$

Actual discharge $Q_a = \underline{\hspace{2cm}} \text{Cu.m/s}$

2) Slope of the channel $= i = \underline{\hspace{2cm}}$

Depth of water level $= H = \underline{\hspace{2cm}} \text{m}$

Width of channel $B = 0.15 \text{ m}$

Area of flow $a = B H = \underline{\hspace{2cm}} \text{sq.m}$

Velocity of flow $V = Q/(B H) = \underline{\hspace{2cm}} \text{m/sec}$

Hydraulic mean radius $m = BH/(B+2H) = \underline{\hspace{2cm}} \text{m}$

by Chezy's formula,

$$V = C \sqrt{mi}$$

Hence, C, Chezy's constant is given by

$$C = \frac{V}{\sqrt{mi}}$$

SET 4 :

1. Area of tank a = 0.6x0.5 sq.m

= 0.3 sq.m

Rise of level = _____m.

Volume collected = _____cu.m.

Time taken = _____secs.

Actual discharge Qa = _____Cu.m/s

2) Slope of the channel = i = _____

Depth of water level = H = _____ m

Width of channel B = 0.15 m

Area of flow a = B H = _____sq.m

Velocity of flow V = Q/(B H) = _____m/sec

Hydraulic mean radius $m = BH/(B+2H) = \text{_____}m$

by Chezy's formula,

$$V = C \sqrt{mi}$$

Hence, C, Chezy's constant is given by

$$C = \frac{V}{\sqrt{mi}}$$

CONCLUSION:

Marks obtained:	Signature of faculty:	Date:
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Date:

Fluid Mechanics-2 (CL372)

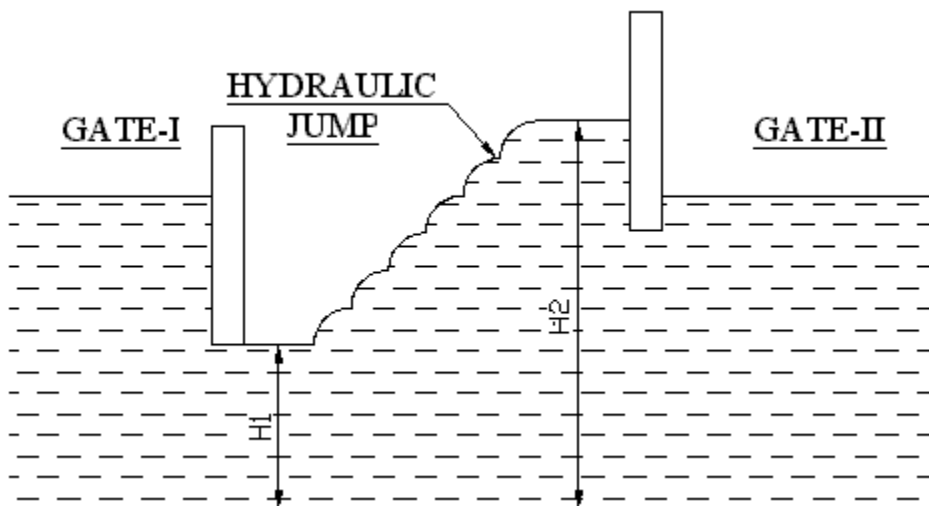
EXPERIMENT NO. 13

TO DETERMINE THE CONJUGATE DEPTH AND LOSS OF SPECIFIC ENERGY IN A HYDRAULIC JUMP

HYDRAULIC JUMP

AIM:

To determine the conjugate depth and loss of specific energy in a Hydraulic Jump for a given slope.



THEORY:

Hydraulic jump is the jump or standing wave formed when the depth of flow of water changes from supercritical to subcritical state.

When the slope of open channel decreases from steep to mild, the depth of flow of water increases toward the critical depth and a flow instability occurs at some point. The flow becomes turbulent until the new normal depth is attained in the downstream. This is called a hydraulic jump.

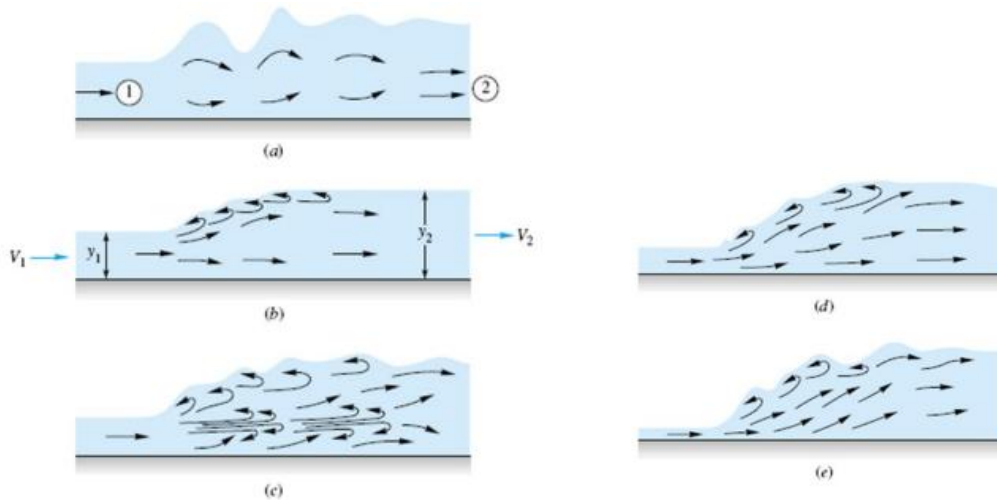
A hydraulic jump occurs when a liquid at a high velocity discharges into a zone that has a lower velocity. The slowing of the liquid leads to an increase in height that changes the kinetic energy of the liquid into potential energy. Some of the energy is dissipated in the form of heat due to turbulence.

Froude Number: $Fr = V/\sqrt{gL}$

Where, V = Velocity g = gravity L = depth of flow

When Froude number greater than 1 is a supercritical flow. When Froude number less than 1 is a subcritical flow. In order to have a hydraulic jump the Froude number needs to be greater than or equal to 1. A hydraulic jump occurs when the flow goes from supercritical flow ($Fr > 1$) to subcritical flow ($Fr < 1$) or from an unstable flow to a stable flow. A hydraulic jump will not occur when a flow goes from subcritical flow ($Fr < 1$) to a supercritical flow ($Fr > 1$).

Types of Jump :



Classification of hydraulic jumps:

- (a) $Fr = 1.0$ to 1.7 : undular jumps;
- (b) $Fr = 1.7$ to 2.5 : weak jump;
- (c) $Fr = 2.5$ to 4.5 : oscillating jump;
- (d) $Fr = 4.5$ to 9.0 : steady jump;
- (e) $Fr = 9.0$: strong jump.

- a. Weak (Undular) Jump : Low energy dissipation rate, Smooth downstream water surface
- b. Oscillating Jump : Irregular fluctuations of flow, Causes turbulence downstream
- c. Steady Jump : Jump forms steadily at same location and is well balanced, Turbulence is confined within the jump
- d. Strong Jump: Large change in depth of the water surface, High energy dissipation rate

EXPERIMENTAL PROCEDURE:

1. Set the channel for a given slope.
2. Remove all the models.
3. Close both the inlet (I) and outlet (II) gates.
4. Allow water in the channel.

5. Open the gate II completely.
6. Open the gate I slightly, so that water flows under the gate in supercritical (shooting) condition.
7. Close the gate II gently, so that it causes obstruction to the shooting flow, and a Hydraulic Jump is formed.
8. Regulate the gate II finely that the Hydraulic jump stays at the middle of the channel.
9. With the help of traveling hook gauge measure the conjugate depths of the flow before and after the Hydraulic Jump, H_1 and H_2 .
10. Measure the actual flow rate with the collecting tank.

CALCULATIONS

1. Area of tank a = 0.6×0.5 sq.m

= 0.3 sq.m

Rise of level = _____ m.

Volume collected = _____ cu.m.

Time taken = _____ secs.

Actual discharge Q_a = _____ Cu.m/s

2. Width of the channel B = 0.15 m

Velocity before the jump $V_1 = Q / (B \times H_1) =$ _____ m/s

Froude no, $Fr_1 = V_1 / \sqrt{gH_1} =$ _____

From Theory

$$H_{2t} = (H_1/2) \left(\sqrt{1 + (8Fr_1^2)} - 1 \right)$$

Actual depth after the jump = H_2 = _____m

Find the theoretical H_{2t} for different H_1 and compare with actual H_2 .

Energy before jump = $H_1 + (V_1^2/2g) =$ _____m

Energy after jump = $H_2 + (V_2^2/2g) =$ _____m

Loss of energy = $(H_1 - H_2) + (V_1^2 - V_2^2)/2g =$ _____m

CONCLUSION:

Marks obtained:	Signature of faculty:	Date:
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